

Pratik Vijay Jadhav

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SUMMARY

ML Systems Engineer specializing in **vision systems** and **RAG/LLM applications**, with experience building and deploying **end-to-end AI systems** from **data pipelines** to **low-latency cloud inference**. Improved **defect detection accuracy** from **34% → 92%**, built **RAG systems processing 50k+ profiles**, and developed **real-time APIs supporting 1,000+ concurrent requests**.

SKILLS

Machine Learning / AI: CNNs, Computer Vision, RAG, LLM Systems, LangChain, DSPy

ML Frameworks: TensorFlow, Scikit-learn, Hugging Face

Backend & APIs: FastAPI, REST APIs, WebSockets, Celery, RabbitMQ

Databases: MongoDB, PostgreSQL, MySQL, FAISS

Infrastructure & Tools: Docker, TensorFlow Serving, MLflow, Vertex AI, Git, GitHub, Selenium, Crawl4AI

Languages: Python, SQL

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Joshnik AI Labs

03/2025 – 02/2026

- Designed a **distributed asynchronous task** processing system handling **1,000+ concurrent requests** using FastAPI, RabbitMQ, Celery, and MongoDB.
- Engineered automated **data extraction pipelines** collecting **30k+ structured real-estate listings** across multiple platforms using Selenium and Crawl4AI.
- Led development of a **50k+ professional profile ingestion pipeline**, powering an LLM-based candidate discovery chatbot backed by **FAISS vector search** with sub-second retrieval.
- Architected a multi-tenant **FastAPI-based Razorpay payment system** enabling direct merchant settlements, implementing **AES-encrypted credential storage**, **webhook signature verification**, and **Dockerized deployment**.
- Built a **real-time reservation system** using React and FastAPI WebSockets with **<100ms booking latency**, implementing **atomic MongoDB capacity controls** and **automated scheduled capacity release** to prevent double bookings.

Software Development Intern, Alemeno Private Limited

04/2024 – 10/2024

- Designed and deployed a **multimodal defect-detection system** combining custom CNNs with LLM-based reasoning for LCD quality control.
- Improved classification accuracy from **34% → 92%** across **15+ defect classes**, reducing false positives by **60%**.
- Constructed and scaled a **10,000+ image defect dataset** using automated labeling pipelines, reducing manual annotation effort by **75%** while maintaining **95%+ label accuracy**.
- Deployed models via **Dockerized TensorFlow Serving on Vertex AI**, enabling **<200ms inference latency** and supporting **50+ daily production quality checks**.
- Increased hard-defect detection accuracy to **70–90%**, reducing manual inspection workload by **40%**.

Machine Learning Intern, Cheslab Private Limited

04/2023 – 06/2023

- Curated and preprocessed a **150k-image dermatology dataset**, building reusable pipelines for cleaning, augmentation, and training.
- Optimized a CNN-based skin-condition classifier achieving **74% accuracy** across **8 disease categories**, integrating the model into a user-facing diagnosis workflow.

AI Research Intern - Agriculture LLM for Smart Farming, Atria University

08/2023 – 02/2024

- Developed a **domain-specific agricultural LLM assistant** using retrieval-augmented generation for crop, soil, and irrigation guidance.
- Implemented a **RAG evaluation pipeline** using a **LLaMA-based judge model**, improving retrieval precision by **35%** and reducing hallucinations by **50%**.

PROJECTS

Multi-Document Conversational AI Agent

- Built a **Retrieval-Augmented Generation (RAG)** system for multi-PDF question answering using **LangChain, Google Gemini, and FAISS**, supporting **multi-collection document corpora** with **Top-K and MMR retrieval**, reducing irrelevant context by **~30–40%**.
- Implemented **page-level extraction analytics**, **grounding-aware prompting with citations**, and a **batch evaluation pipeline with JSON reporting**, achieving **>85% citation coverage** and low-latency retrieval.

Facial Emotion-Based Music Recommendation 

- Built a real-time facial emotion recognition system using **CNNs** trained on **FER-2013**, achieving **85% accuracy** across **7 emotion classes**.
- Optimized inference to maintain **<150ms latency** per frame and launched an interactive **Streamlit application** integrated with the **Spotify API**.

EDUCATION

B.Tech in Computer Science (Digital Transformation), Atria University

11/2021 – 08/2025 | Bangalore, Karnataka

- CGPA: 9.02